25901405

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC)-MG4SECMBG200

SOLID WASTE MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A - Short Notes Questions - Answer any 10 questions- ***Each question carries 2 marks***

1. Define open dumping. **[K] / [CO1]**

Open dumping is the practice of disposing of solid waste in an unregulated and uncontrolled manner, where waste is simply thrown onto land or into water bodies without any proper treatment, management, or environmental protection measures. 2marks

2. Name two agricultural wastes. **[K] / [CO1]**

Examples of agricultural waste include: Rice husk , Wheat straw, Sugarcane bagasse , Corn cobs , Coconut shells, Banana peels and leaves , Groundnut shells, Animal manure (cow dung, poultry litter). Any two 2 marks

3. List out factors affecting composting. **[K] / [CO2]**

Temperature , Moisture content , Oxygen supply (aeration) , Carbon to nitrogen (C: N) ratio, pH, Particle size , Microorganisms , Turning/mixing of compost. Any four factors, 2 marks

4. Why are epigeic earthworms preferred for vermicomposting? **[K] / [CO2]**

Surface dwellers, High feeding rate, Fast reproduction, Efficient decomposition, Tolerance to organic waste conditions. Any two points 2 marks

5. What is bed method of vermicomposting? **[K] / [CO2]**

The bed method of vermicomposting is a technique in which organic waste is decomposed by earthworms in raised beds above the ground surface.

6. What is cold composting? **[U] / [CO2]**

Cold composting is a slow, natural method of composting in which organic waste decomposes without maintaining high temperatures or frequent turning.

7. Which types of polymers are broken down during hydrolysis? **[K] / [CO2]**

Polymers that are broken down during hydrolysis are mainly biodegradable polymers containing hydrolysable bonds. Examples include: Carbohydrates (e.g., starch, cellulose), Proteins, Lipids (fats and oils), Nucleic acids.

8. Explain why soil is added during tube composting. [U] / [CO3]

Soil is added during tube composting for the following reasons: Source of microorganisms, provides enzymes, Odour control, maintains moisture balance, and improves compost quality.

9. State one advantage of in-vessel composting. [K] / [CO3]

Advantages of in-vessel composting include: Faster composting process due to controlled conditions. Better control of temperature, moisture, and aeration. Reduced odour and pest problems, requires less space compared to open methods, can be carried out in all weather conditions, produces uniform and high-quality compost, and minimises environmental pollution. Anyone 2 marks

10. What is compost? [K] / [CO3]

Compost is a dark, nutrient-rich, humus-like material formed by the decomposition of organic waste (such as plant residues, kitchen waste, and animal manure) by microorganisms under controlled aerobic conditions.It is commonly used as a natural fertiliser and soil conditioner to improve soil fertility, structure, and water-holding capacity. 2 marks

11. Name any two microorganisms involved in open pile composting. [K] / [CO3]

Any two microorganisms, 2 marks

12. Name two byproducts formed during incineration? [K] / [CO1]

Ashes, gases and heat

Part B

Short Essay Questions

Answer any 4 questions

Each question carries 5 marks

13. Describe how microorganisms help in pit composting. [K] / [CO3]

Microorganisms play a key role in pit composting by breaking down organic waste into compost. Decomposition of organic matter, Enzymatic action, Heat production, Humus formation, Nutrient recycling.

14. Explain the need for sampling in solid waste management. [U] / [CO1]

Sampling in solid waste management is important to obtain a representative portion of waste for analysis and planning. Determines waste composition, Estimates quantity and characteristics, Aids in selecting treatment methods, supports system design, Cost estimation, Monitoring and evaluation.

15. List the advantages of biogas as a waste disposal method. [K] / [CO1]

Reduces organic waste volume by converting it into useful products - Produces renewable energy (biogas) used for cooking, heating, and electricity - Generates nutrient-rich slurry that can be used as organic fertiliser - Environment-friendly – reduces greenhouse gas emissions and pollution.Controls odour and pathogens during digestion - Reduces dependence on fossil fuels, Low-cost and sustainable for rural and household use

16. Describe the environmental significance of waste reduction. [K] / [CO1]

Conserves natural resources , reduces pollution, decreases landfill use , lowers greenhouse gas emissions , Saves energy , protects ecosystems and biodiversity , promotes sustainability .

17. Explain in detail the end products of composting. [U] / [CO2]

Humus (finished compost) – A dark, crumbly, soil-like material rich in nutrients; improves soil fertility, structure, and water-holding capacity. Carbon dioxide (CO_2) – Released during microbial respiration. Water (H_2O) – Produced during decomposition and helps maintain moisture. Heat – Generated due to microbial activity, which aids in pathogen destruction. Mineral nutrients – Such as nitrogen, phosphorus, potassium, and trace elements in forms available to plants.

18. Explain the composition and properties of biogas. [U] / [CO2]

Composition of biogas: Methane (CH_4), Carbon dioxide (CO_2), Hydrogen sulphide (H_2S), Hydrogen (H_2), Nitrogen (N_2), and water vapour. Properties of biogas: Combustible gas due to methane content, Colourless and odourless (pure form; odour due to H_2S), produces smokeless flame when burned, Moderate calorific value, Lighter than air Eco-friendly fuel, reduces pollution and greenhouse gas emissions.

Part C
Essay Question
Answer any 1 questions
Each question carries 10 marks

19. Describe the advantages and disadvantages of composting. [K] / [CO2]

Advantages: Reduces waste volume and decreases landfill burden, Produces nutrient-rich compost that improves soil fertility, Environment-friendly – reduces pollution and greenhouse gas emissions, Improves soil structure and water-holding capacity, Recycles organic waste into useful material, Low-cost and simple method for waste management, Reduces need for chemical fertilisers.[**5]** Disadvantages: Slow process – takes weeks to months, requires proper maintenance (moisture, aeration, temperature), may produce foul odour if not managed properly. Attracts pests and insects, requires space for a composting setup, Not suitable for all types of waste (e.g., plastics, metals).**[5]**

20. Describe the structure and preparation of a compost bin. [K] / [CO3]

Structure & Types of a Compost Bin: Container/Enclosure – Made of plastic, wood, metal, or concrete Base – Usually open or with drainage to allow excess water to escape and contact with soil organisms Lid/Cover – Layers inside – Alternate layers of organic waste (greens) and dry materials (browns) [**5].** Preparation of a Compost Bin: [**5]**